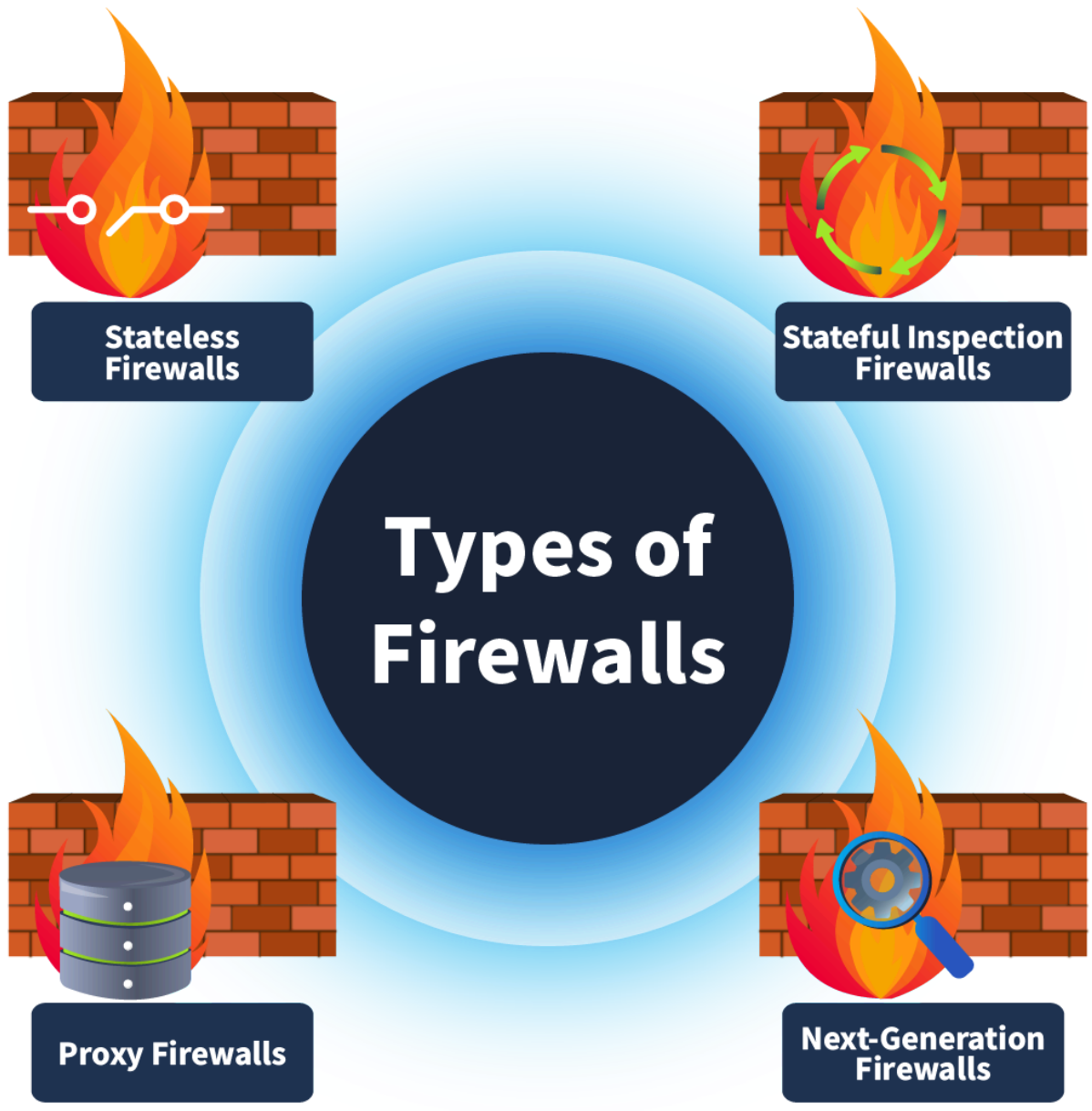
# Firewall Fundamentals

A firewall is designed to inspect a network's or digital device’s incoming and outgoing traffic. The firewall will allow or deny that traffic based on its maintained rules.

# Types of Firewalls

different types of firewalls work on different OSI model layers




## Stateless Firewall

operates on layer 3 and layer 4 of OSI model and works by filtering data based on predetermined rules without taking note of previous connections.  This means it will match every packet with the rules regardless of whether it is part of a legitimate connection.

## Stateful Firewall

this type of firewall goes beyond filtering packets by predetermined rules.  It also keeps track of previous connections and stores them in a state table. This adds another layer of security by inspecting the packets based on their history with connections. Stateful firewalls operate at layer 3 and layer 4 of the OSI model.

## Proxy Firewall

Proxy firewalls, or application-level gateways, act as intermediaries between the private network and the Internet and operate on the OSI model's layer 7.

They inspect the content of all packets as well. The requests made by users in a network are forwarded by this proxy after inspection and masking them with their own IP address to provide anonymity for the internal IP addresses

## Next-Generation Firewall (NGFW)

firewall that operates from layer 3 to layer 7 of the OSI model, offering deep packet inspection and other functionalities that enhance the security of incoming and outgoing network traffic. It offers heuristic analysis by analysing the patterns of attacks and blocking them instantly before reaching the network.

| Firewalls | Characteristics |
|---|---|
| Stateless firewalls | - Basic filtering<br>- No track of previous connections<br>- Efficient for high-speed networks |
| Stateful firewalls | - Recognize traffic by patterns<br>- Complex rules can be applicable<br>- Monitor the network connections |
| Proxy firewalls | - Inspect the data inside the packets as well<br>- Provides content filtering options<br>- Provides application control<br>- Decrypts and inspects SSL/TLS data packets |
| Next-generation firewalls | - Provides advanced threat protection<br>- Comes with an intrusion prevention system<br>- Identify anomalies based on heuristic analysis<br>- Decrypts and inspects SSL/TLS data packets |

# Rules in Firewalls

The basic components of a firewall's rule are described below:

- **Source address:** The machine's IP address that would originate the traffic.
- **Destination address:** The machine's IP address that would receive the data.
- **Port:** The port number for the traffic.
- **Protocol:** The protocol that would be used during the communication.

- **Action:** This defines the action that would be taken upon identifying any traffic of this particular nature.
- **Direction:** This field defines the rule's applicability to incoming or outgoing traffic.

## Types of Actions

The component "Action" from a rule indicates the steps to take after a data packet falls under the category of the defined rule

### Allow

A rule's "Allow" action indicates that the particular traffic defined inside the rule would be permitted.

| Action | Source | Destination | Protocol | Port | Direction |
|---|---|---|---|---|---|
| Allow | 192.168.1.0/24 | Any | TCP | 80 | Outbound |

### Deny

A rule's "Deny" action means that the traffic defined inside the rule would be blocked and not permitted.

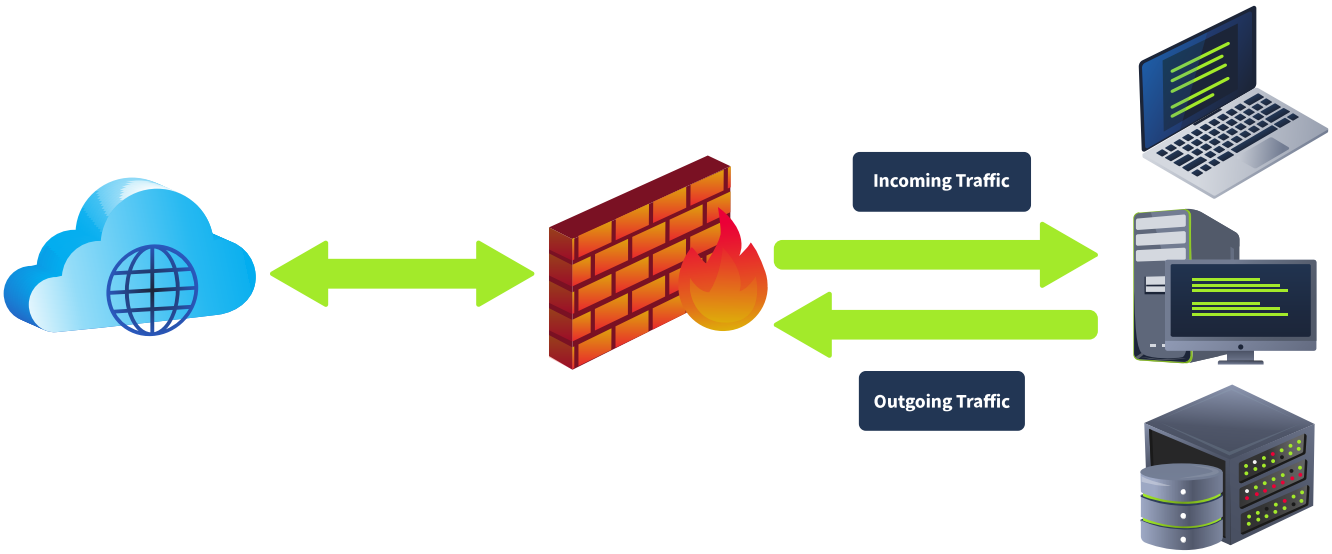
| Action | Source | Destination | Protocol | Port | Direction |
|---|---|---|---|---|---|
| Deny | Any | 192.168.1.0/24 | TCP | 22 | Inbound |

### Forward

The action "Forward" redirects traffic to a different network segment using the forwarding rules created on the firewalls.

| Action | Source | Destination | Protocol | Port | Direction |
|---|---|---|---|---|---|
| Forward | Any | 192.168.1.8 | TCP | 80 | Inbound |

## Directionality of Rules




### Inbound Rules

Rules are categorised as inbound rules when they are meant to be applied to incoming traffic only. For example, you might allow incoming HTTP traffic (port 80) on your web server.

### Outbound Rules

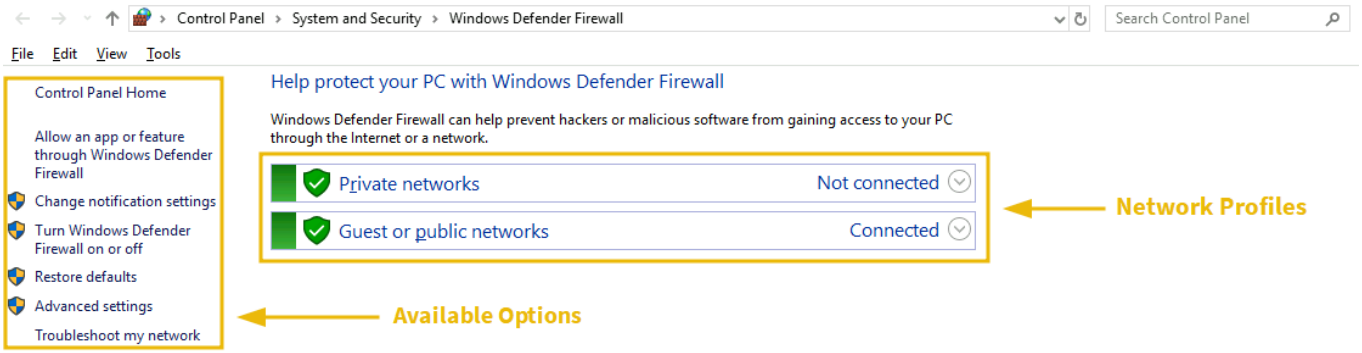
These rules are made for outgoing traffic only. For example, blocking all outgoing SMTP traffic (port 25) from all the devices except the mail server.

### Forward Rules

Forwarding rules are created to forward specific traffic inside the network. For example, a forwarding rule can be created to forward the incoming HTTP (port 80) traffic to the web server located in your network.

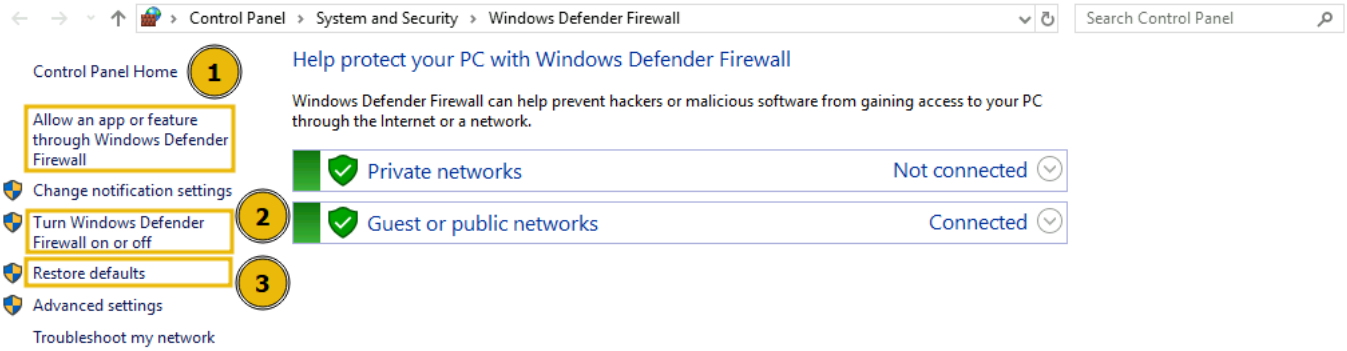
## Windows Defender Firewall

Windows Defender is a built-in firewall introduced by Microsoft in the Windows OS.
This firewall contains all the basic functionality for creating, allowing, or denying specific programs or creating customised rules.



## Network Profiles

1. **Private networks:** This includes the firewall configurations to apply when connected to our home network.
2. **Guest or public networks:** This includes the firewall configurations to apply when connected to a public or untrusted network like coffee shops, restaurants, or similar.



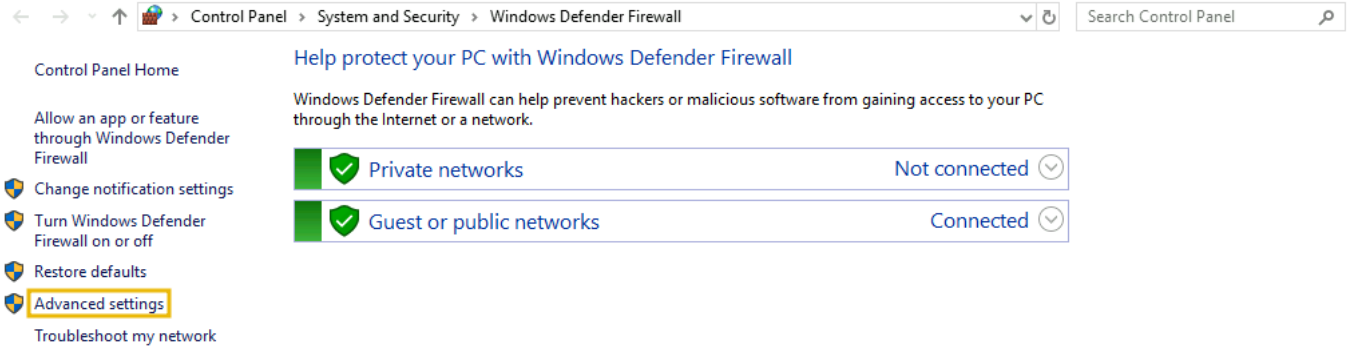

To allow/disallow any application in any of your network profiles, click on the option (highlighted as 1 in the screenshot)

if you want to turn it on/off, you can click on the option (highlighted as 2 in the screenshot).

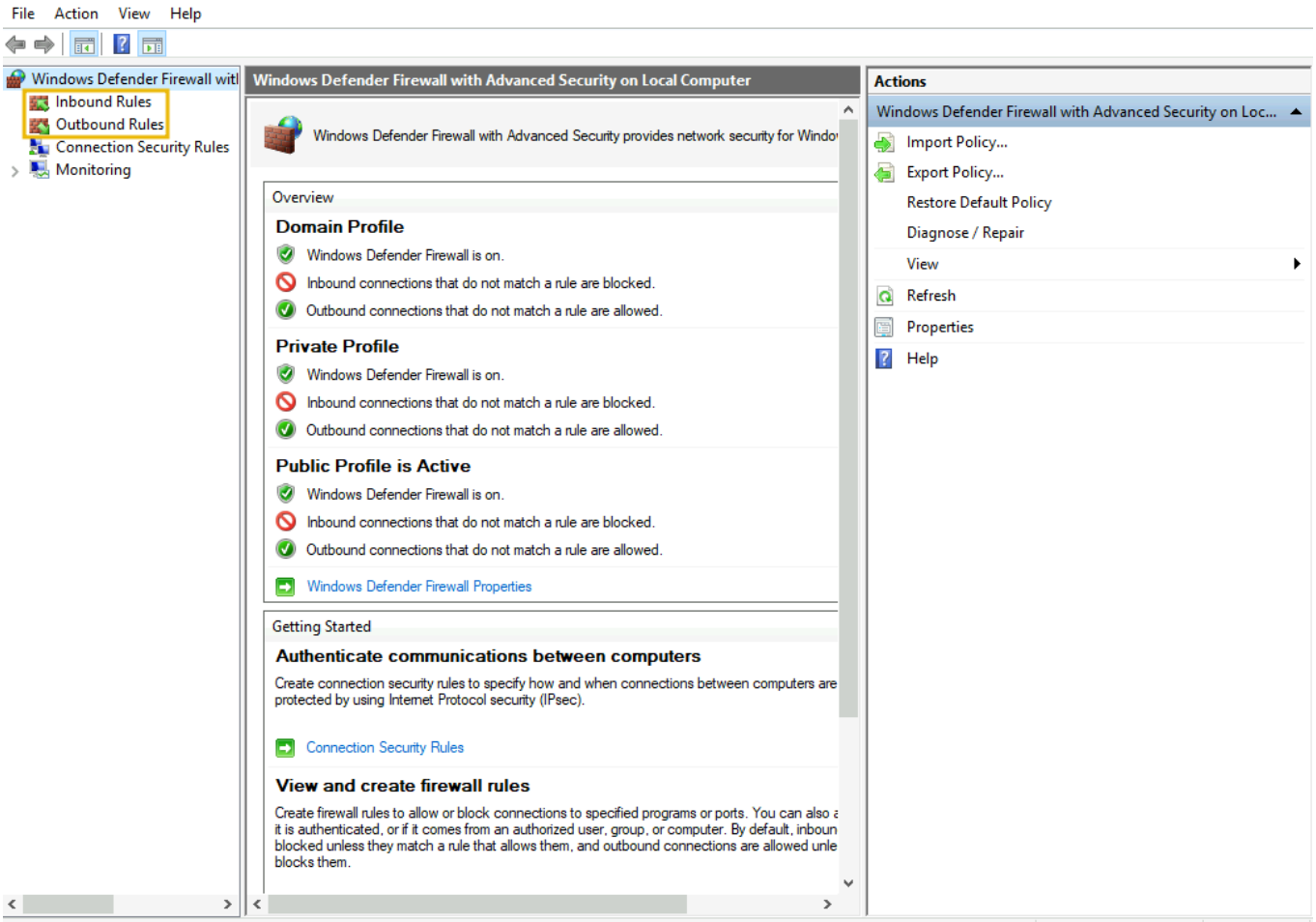
click on "Restore Defaults" (highlighted as 3 in the screenshot) from the main dashboard anytime to restore all the firewall's default settings.

## Custom Rules

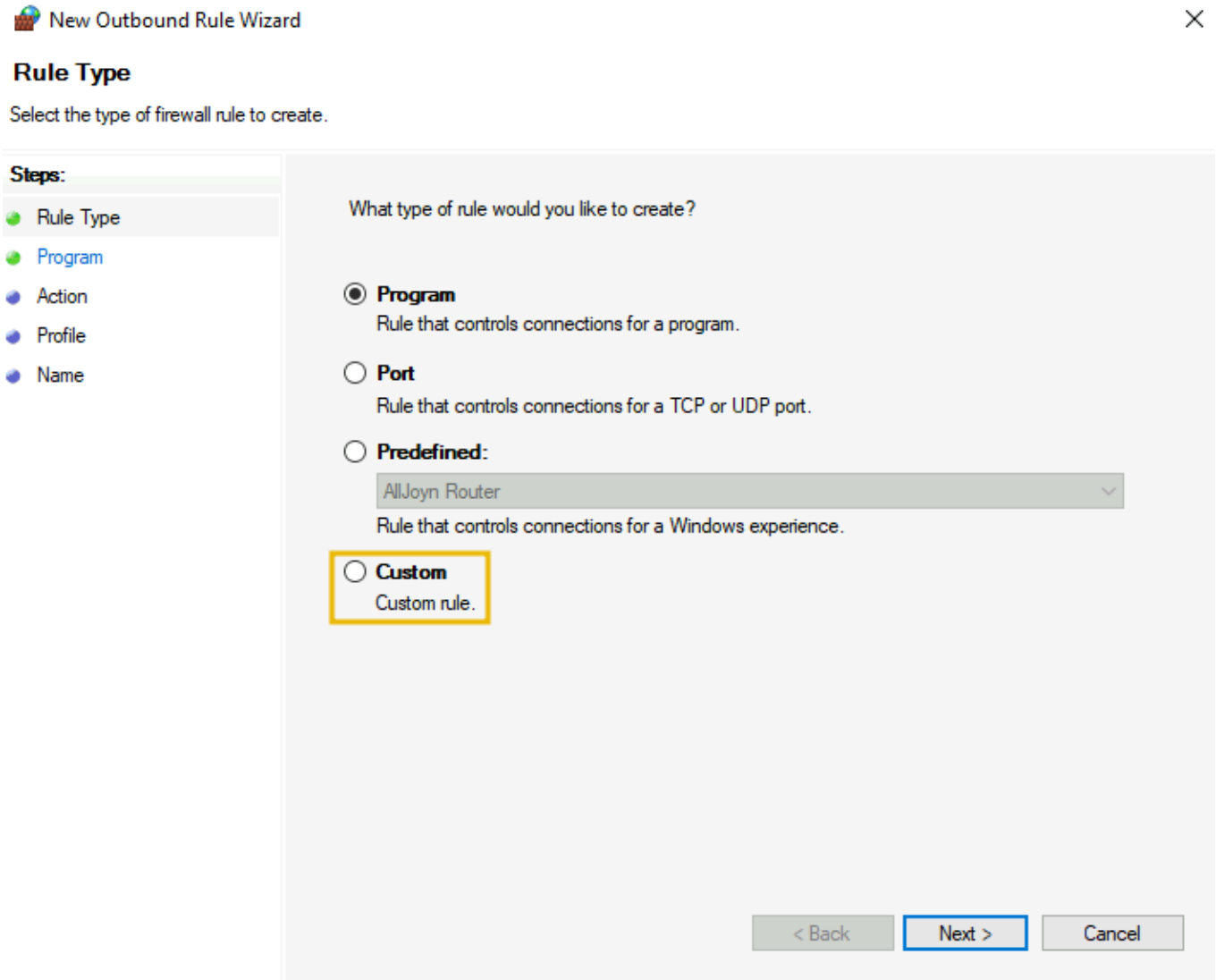
Windows Defender Firewall also allows you to create custom rules for your network to allow/disallow specific traffic as needed



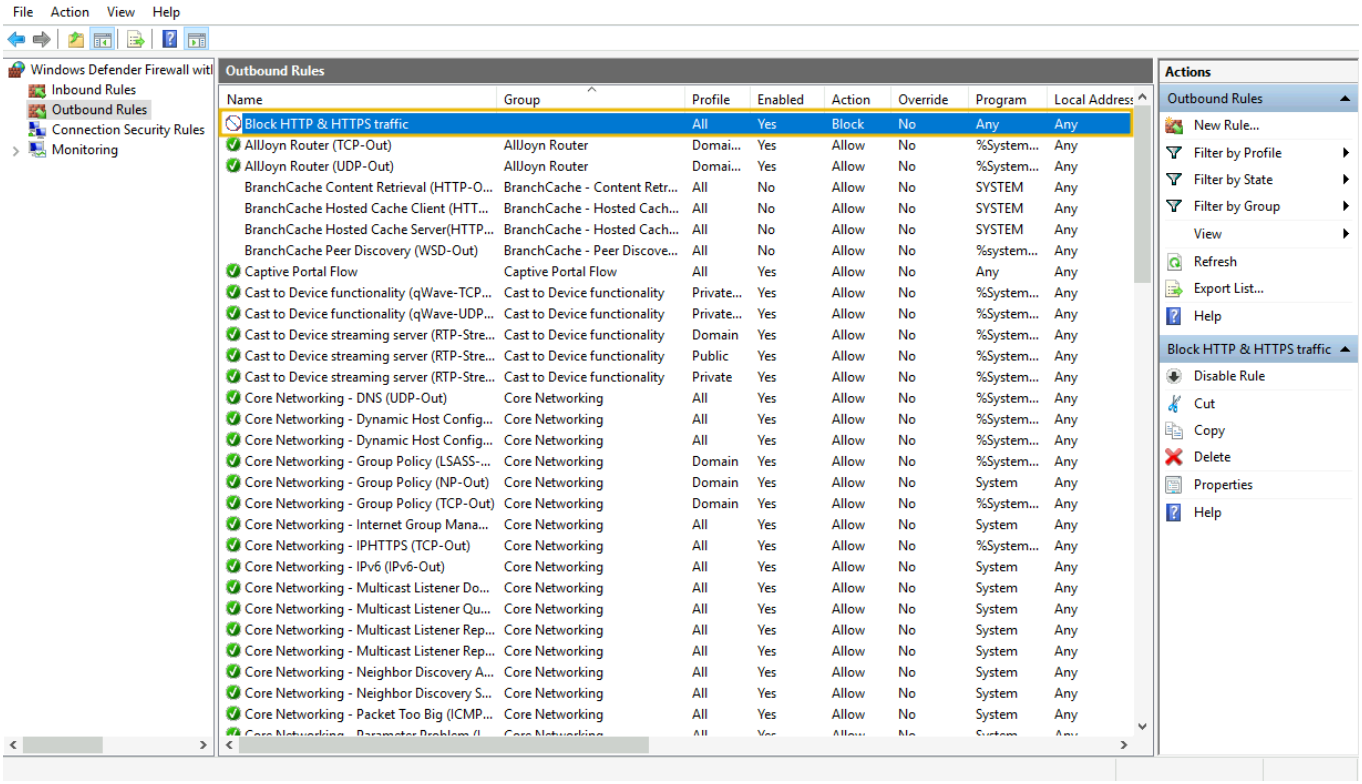
You can see the available options to create inbound and outbound rules.



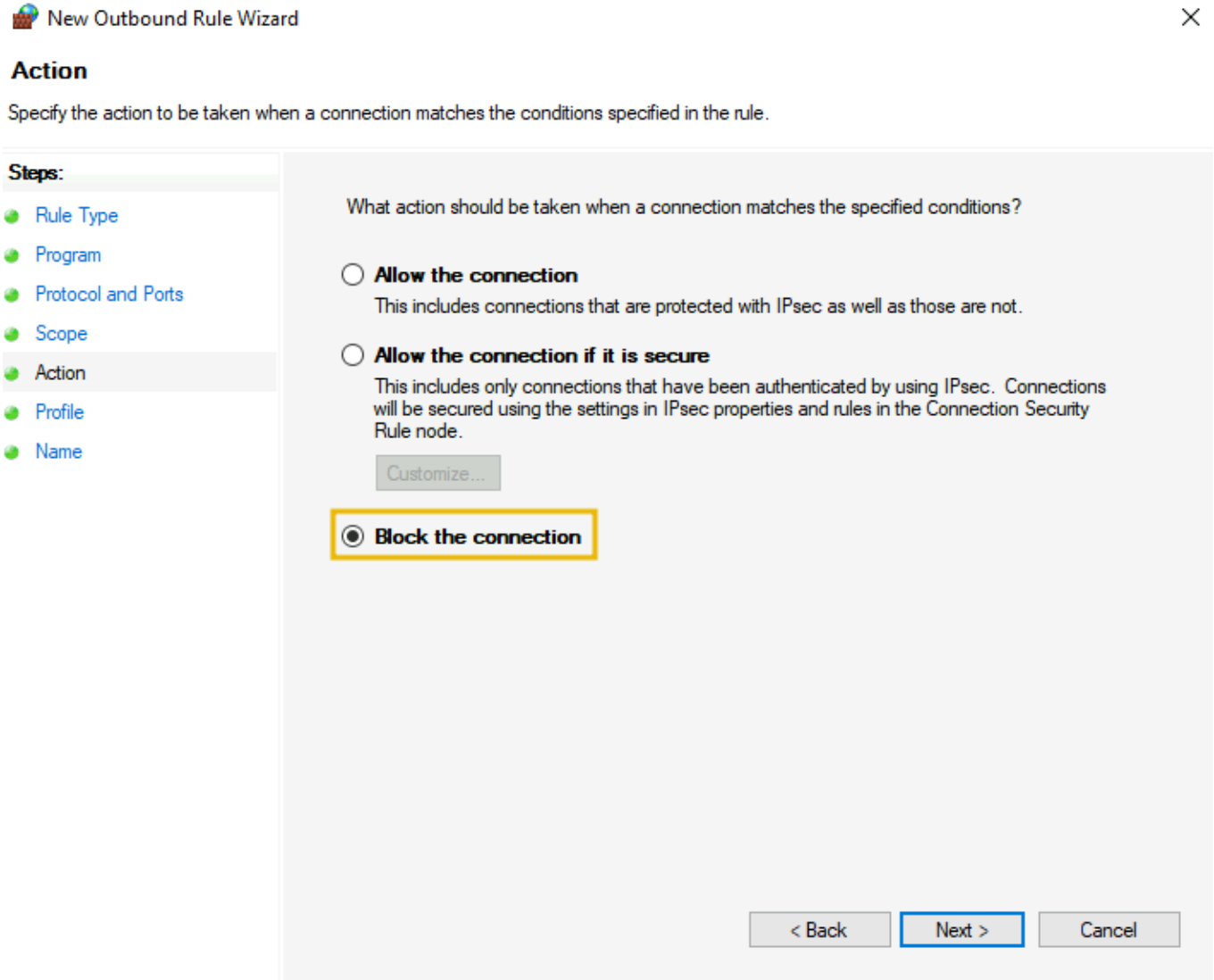

Let’s create an outbound rule to block all our outgoing HTTP and HTTPS traffic. For this, click on the **Outbound Rules** option on the left side, then click on **New Rule** on the right side. It will open the rule wizard.In the first step, select the **Custom** option and press **Next**.



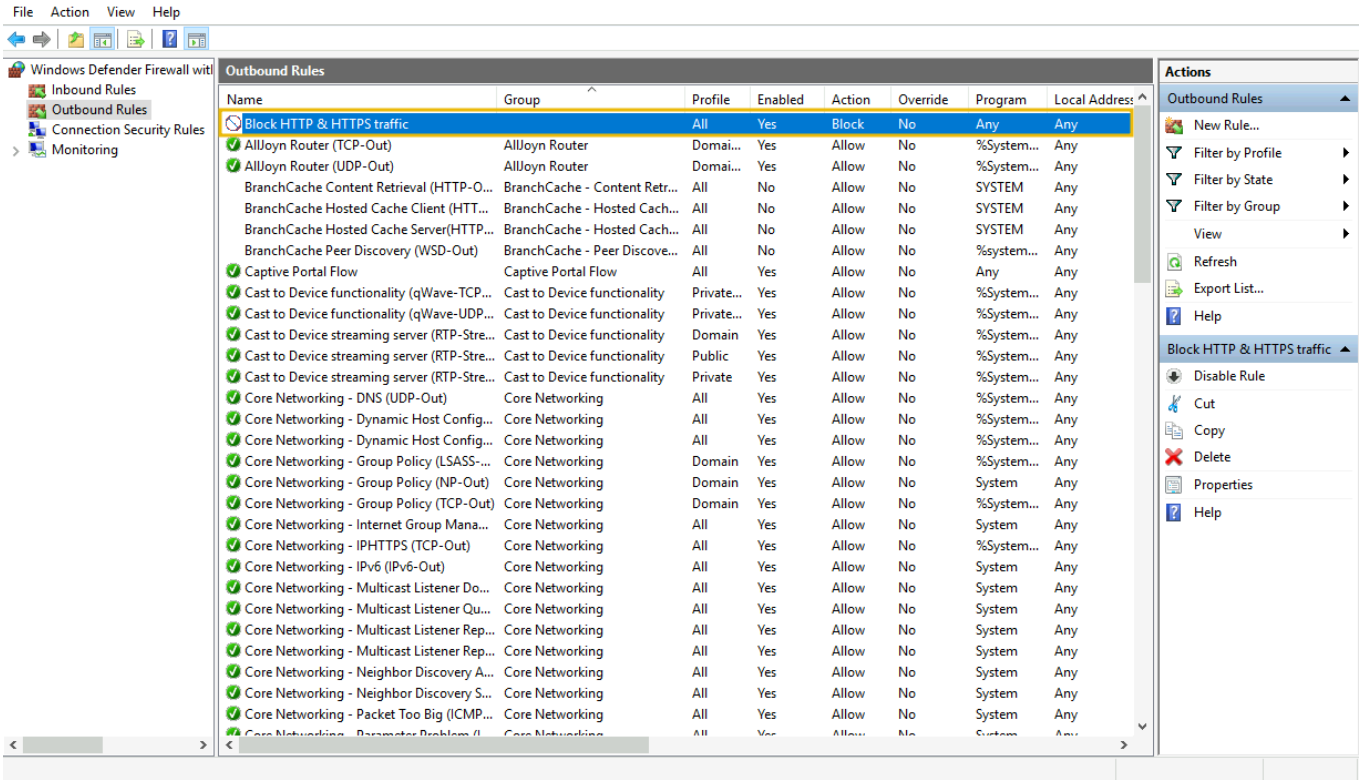

In the second step, select **All programs** from the next option and press **Next**. It will ask you to select the protocol type in the third step. Select the **Protocol type** as "TCP", keep the **Local port** as it is, and change the **Remote port** to "Specific ports" from the dropdown




In the **Scope** tab, keep the local and remote IP addresses as they are and press the **Next** button. In the **Action** tab, enable the **Block the connection** option and press **Next**.




In the **Profile** tab, we keep all the network profiles check-marked. Lastly, the final phase is to give your rule a name and an optional description and press the **Finish** button.




# Linux iptables Firewall

## Netfilter

Netfilter is the framework inside the Linux OS with core firewall functionalities, including packet filtering, NAT, and connection tracking

- **iptables:** This is the most widely used utility in many Linux distributions. It uses the Netfilter framework that provides various functionalities to control network traffic.
- **nftables:** It is a successor to the “iptables” utility, with enhanced packet filtering and NAT capabilities. It is also based on the Netfilter framework.
- **firewalld:** This utility also operates on the Netfilter framework and has predefined rule sets. It works differently from the others and comes with different pre-built network zone configurations.

## ufw

ufw (Uncomplicated Firewall), as the name says, eliminates the complications of making rules in a complex syntax in “iptables”(or its successor) by giving you an easier interface

To check the status of the firewall, you could use the command below:

Check status

```
user@ubuntu:~$ sudo ufw status
Status: inactive
```

If it appears inactive, you can enable it using the following command:

Enable firewall

```
user@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

To turn off the firewall, type “disable” instead of “enable” in the above command.

Below is a rule created to allow all the outgoing connections from a Linux machine.

Allow all outgoing traffic

```
user@ubuntu:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
```

`default` in the command means that we are defining this policy as a default policy allowing all the outgoing traffic unless we define an outgoing traffic restriction on any specific application in a separate rule.

You can deny incoming traffic at any port in your system

We can achieve this with the command `ufw deny 22/tcp` - this would deny connections from tcp port 22

Deny incoming SSH traffic

```
user@ubuntu:~$ sudo ufw deny 22/tcp
Rule added
Rule added (v6)
```

To list down all the active rules in a numbered order, you can use the following command:

List all rules

```
user@ubuntu:~$ sudo ufw status numbered
     To                         Action      From
     --                         ------      ----
[ 1] 22/tcp                     DENY IN     Anywhere
[ 2] 22/tcp (v6)                DENY IN     Anywhere (v6)
```

To delete any rule, execute the following command with the rule number to delete:

Delete a rule

```
user@ubuntu:~$ sudo ufw delete 2
Deleting:
 deny 22/tcp
Proceed with operation (y|n)? y
Rule deleted (v6)
```